

- 1 The law which represents the Boolean equation $A + B = B + A$ is:
- Associative Law
 - Distributive Law
 - Commutative Law
 - Absorption Law

- 2 The dual of the Boolean equation $(X+Y) \cdot Z = X \cdot Y \cdot Z$ is:
- $X+Y+Z$
 - $X \cdot Y + Z$
 - $(X \cdot Y) + Z$
 - $(X+Y) + Z$

- 3 If A=1, B=0, C=0 and D=1, then the minterm will be:

- $ABCD$
- $A'BC'D$
- $A'B'C'D$
- $A'B'C'D'$

- 4 The complement of the Boolean expression $F(P,Q,R) = (P+Q+R)'$ is:
- $P'Q'R'$
 - $P'+Q'+R'$
 - $P+Q+R$
 - $(P+Q) \cdot R'$

- 5 The propositional operator $\Rightarrow$ represents:

- Conjunction
- Implication
- Disjunction
- Negation

- 7 NAND gate is formed by the combination of:

- AND gate and OR gate
- OR gate and NOT gate
- NAND gate and NOT gate
- AND gate and NOT gate

- 10 The proposition $(a \leftrightarrow b)$ is represented by:

- $(a+b) \cdot (a+b)$
- $(a+b) \cdot (a+b)$
- $(a+b)'$
- $(a+b)$

- 11 If the input in a logic is $ABCD$, then the decimal equivalent output will be:

- 8
- 10
- 5
- 6

- 13 With reference to the given proposition $\neg P \Rightarrow Q$, answer the following questions:

- the converse of the proposition is:
- $\neg Q \Rightarrow P$
- $Q \Rightarrow P$
- $\neg Q \Rightarrow \neg P$
- $\neg P \Rightarrow \neg Q$

- 14 the contrapositive of the proposition is:

- $\neg P \Rightarrow \neg Q$
- $Q \Rightarrow P$
- $\neg Q \Rightarrow P$
- $\neg P \Rightarrow \neg Q$

- 16 What is the output of the statement given below?

System.out.println("2345" < "A");

- 2345-05
- 2345
- 230
- ERROR

- 17 What is the output of the statement given below?

System.out.println("A" < "1" < "C");

- 05-1-06
- 10-1-07
- 081
- 05-05-07

- 18 The basic logic gate that represents the simplification of the Boolean expression $A(A+B)(A+B)$ is:

- OR gate
- NOT gate
- AND gate
- None of the above

- 20 The proposition $\neg(a \wedge b) \vee (a \wedge b)$ is a:

- Contradiction
- Contingency
- Tautology
- Implication

- 22 A school intends to select candidates for an inter school competition as per the criteria given below:

- The student has participated in an earlier competition and is very creative.
- The student is very creative and has excellent general awareness, but has not participated in any competition earlier.
- The student has excellent general awareness and has won prize in an inter school competition.

The inputs are:

Inputs	
A	Participated in a competition earlier
B	Is very creative
C	Won prize in an inter school competition
D	Has excellent general awareness

(In all the above cases 1 indicates yes and 0 indicates no).

Output: X (1 indicates yes and 0 indicates no for all cases).

Draw the truth table for the inputs and outputs given above and answer the following questions:

X	A	B	C	D
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1

- [1]

A	B	XOR
0	0	0
0	1	1
1	0	1
1	1	0

SOP of $XOR(A,B) = \bar{A}B + A\bar{B}$

POS of $XOR(A,B) = (A+B)(\bar{A}+\bar{B})$

$= \bar{A}B + A\bar{B}$

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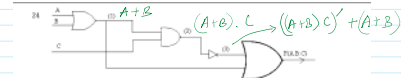
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2	0	0	1	1	✓
3	0	0	0	0	✓
4	0	1	0	0	✓
5	0	1	1	0	✓
6	0	1	1	1	✓
7	1	0	0	0	✓
8	1	0	0	1	✓
9	1	0	1	0	✓
10	1	0	1	1	✓
11	1	1	0	0	✓
12	1	1	0	1	✓
13	1	1	1	0	✓
14	1	1	1	1	✓
15	1	1	1	1	✓



From the logic diagram given above, where A, B and C are inputs and F is the output, answer the following questions.

(a) The expression at (1) is: [1]

(i) $A \cdot B$

(ii) $A \cdot B'$

(iii) $A \cdot B'$

(iv) $A \cdot B$

(b) The expression at (2) is: [1]

(i) $A \cdot B \cdot C$

(ii) $(A+B) \cdot C$

(iii) $(A+B) \cdot C'$

(iv) $(A+B) \cdot C$

(c) The final expression $F(A, B, C)$ is: [1]

(i) $F = (A+B) \cdot C' + (A+B)$

(ii) $F = (A+B) \cdot C' + A \cdot B$

(iii) $F = (A+B) \cdot C' + (A+B)$

(iv) $F = (A+B) \cdot C' + (A+B)$

25 Given the Boolean expression $F = (P + Q)' \cdot (P \cdot Q + Q \cdot R)$, identify:

(a) The complement of the expression: [2]

(i) $P' \cdot R' + (P' \cdot Q') \cdot (Q' \cdot R)$

(ii) $(P' \cdot R') \cdot (P' \cdot Q') \cdot (Q' \cdot R)$

(iii) $P' \cdot R' \cdot (P' \cdot Q') \cdot (Q' \cdot R)$

(iv) $(P' \cdot R') \cdot (P' \cdot Q') \cdot (Q' \cdot R)$

(b) The law used: [1]

(i) Distributive Law

(ii) De Morgan Law

(iii) Associative Law

(iv) Idempotent Law

Question 1

(a) Absorption law states that:

(i) $A \cdot (A' + B) = A$

(ii) $A + (A \cdot B) = A$

(iii) Both (i) and (ii)

(iv) $A \cdot (B + C) = A \cdot B + A \cdot C$

(b) Assertion: A=0 B=1 C=0 and D=1 and minimum is $A \cdot B \cdot C \cdot D$ Reason: The final sum term must be 0 so A and C are complemented. [1]

Which one of the following options is correct?

(i) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(ii) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(iii) Assertion is true and Reason is false.

(iv) Assertion is false and Reason is true.

(c) According to the Principle of duality, the Boolean equation $(P + Q) \cdot R = 1 + P \cdot R + Q \cdot R$ will be equivalent to: [1]

(i) $P + Q + R + 1 + (P \cdot R) + (Q \cdot R)$

(ii) $P + Q + R + 0 + (P \cdot R) + (Q \cdot R)$

(iii) $P + Q + R + 1 + (P \cdot R) + (Q \cdot R)$

(iv) $P + Q + R + 0 + (P \cdot R) + (Q \cdot R)$

(d) The complement of the Boolean expression $(X + Y)' \cdot Z'$ is: [1]

(i) $(X + Y) \cdot Z$

(ii) $X \cdot Y \cdot Z$

(iii) $(X' + Y') \cdot Z'$

(iv) $(X' + Y') \cdot Z$

(e) The equivalent of $P \cdot Q \vee \neg P \wedge \neg Q$ will be: [1]

(i) $(P \wedge Q) \vee (P \wedge \neg Q)$

(ii) $(P \wedge Q) \vee (\neg P \wedge \neg Q)$

(iii) $P \wedge Q \vee \neg P \wedge \neg Q$

(iv) $P \wedge Q \vee (\neg P \wedge \neg Q)$

(f) Assertion: Boolean algebra and Binary number systems are different from each other. Reason: There are some basic operations like AND, OR and NOT which are performed only in Boolean algebra. [1]

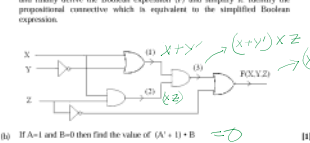
(i) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(ii) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(iii) Assertion is true and Reason is false.

(iv) Assertion is false and Reason is true.

(g) From the logic circuit diagram given below, name the outputs (1), (2) and (3) and finally derive the Boolean expression (P) and simplify it. Identify the propositional connective which is equivalent to the simplified Boolean expression. [4]



(a) If A=1 and B=0 then find the value of $(A' + 1) \cdot B$

Question 5

(i) Draw the logic circuit to encode the following hexadecimal number: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F to its binary counterparts. Also state the binary equivalents of [5]

(a) If $A=1$ and $B=0$ then find the value of $(A' + 1) \cdot B = 0$

(b)
$$= XZ' + Z'$$

$$= (Z' + X)(Z' + Z)$$

$$= (Z' + X)$$

 It is implication meaning 'if then'.

Question 5

(i) Draw the logic circuit to encode the following Hexadecimal number (1, 3, 5, 6, 8, A, C, E) to its binary equivalents. Also state the binary equivalents of the given numbers.

(b)

(ii) Verify if the following proposition is valid using the truth table:

$$A \leftrightarrow (B \wedge C) \equiv [A \leftrightarrow B] \wedge [B \Rightarrow C]$$

(b)

0 0 0 0
 0 0 1 1
 0 1 0 1

